



Erbium Minion Isotope TRM

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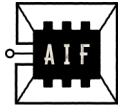
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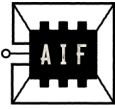
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Contents

Overview	4
Features	5
Signals	7
Test Signals	7
xSPI Target Interface	7
UART Target Interface	7
I2C Controller Interface	7
QSPI Controller Interface	8
GPIO Interface.	8
JTAG Signals	8
Pin Muxing	8
Bring up sequence.	10
First boot on the tester.	10
Second Boot.	11
Program Mode	11
Application Code	11
Power Reset & Clock Management.	12
Clock and reset signals.	12
System Clocks	12
OSC Clock	13
xSPI Clock	14
CPU Subsystem External Resets	14
Reset Sequencing	15
Valid Power State	16
Power State Controller FSM.	17
Interrupts	20
CPU Handling of system interrupts	21
Interrupt handling process.	21



Memory Map	23
Interconnect: Transaction Initiators and Targets	23
CPU and UART Memory Map.	24
xSPI Memory Map.	24
XSPI Datasheet	26
Commands	26
Default Mode.	28
Not supported in Phase 1	29
Verification Requirement.	29
Bootng.	29
Error handling.	30
Errata:	30
SFDP_Reg address map	31
SCCR_Reg address map	79
Steps for running verification	83



Overview

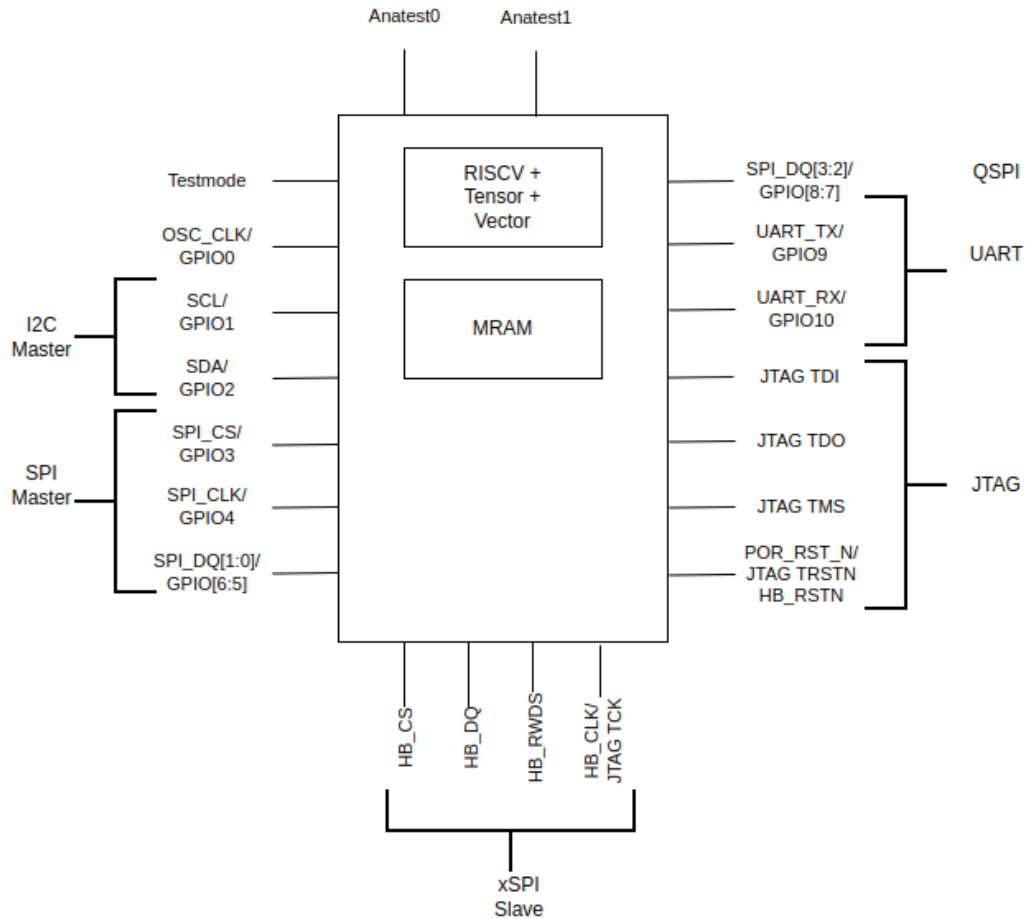
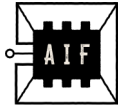


Figure 1: Erbuim RISC-V Isotope

Erbium is

- 8C,16T RISC-V Core, ISA(RV64IMFC_Zicsr_Zifencei +Tensor, +Vector +Atomic)
- I2C and QSPI Controllers
- xSPI and UART Target interfaces
- 16MB MRAM (L2) with 16KB of OTP region with 12KB of user usable OTP's
- Multiple Boot Options
 - Boot from Bootrom.



- Boot from MRAM. (via OTP setting)
- Boot from SRAM (in test mode)

Features

- External Interfaces
 - JTAG
 - UART
 - xSPI Target,
 - I2C Controller
 - (Q)SPI Controller
 - 11 GPIO
 - Chiplet Interfaces
 - ★ AHB
 - ★ AXI

xSPI Interface

- JESD251C and JESD216H compliant
- Supports
 - 1S-1S-1S
 - 4S-4D-4D
 - 4D-4D-4D
 - 8S-8S-8S
 - 8D-8D-8D (Octal)
 - 8D-8D-8D (Hyperbus)
- Operating frequency 200 Mhz
- 64 bit register and memory access
- Linear Burst capable.

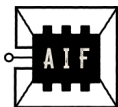
Chiplet Interface

- 2 Interfaces AXI, AHB
- AXI and AHB in 32bit and 64 bit variants.
- Chiplet interface is not available in testchip tapeouts.

Note: While the chiplet interface is a part of the design it is not available (optimised away during PD) in testchips. i.e. erbium will not have the chiplet interface.

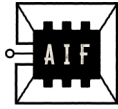
CPU Subsystem

- ET-Minion
 - RV64IMFC + Zicsr + Zifencei
 - In-Order execution with 2 Hardware Threads
- 8 ET-Minions
- L1 Instruction cache
 - Shared among the 8 ET-Minions
 - 128 sets, 4 way (set associative)
 - 64Byte cache line
- L0 Micro Instruction cache
 - Shared among 4 ET-Minions
 - 1 set, 16 ways (fully associative)
- Cooperative Tensor Load
 - Coalesces multiple Tensor Load requests to the same memory location (from different minions) into a single request to provide better performance and power.
- L1 Data Cache (**L1D\$**) - 4KB configurable



CONTENTS

- 16 sets, 4 ways, 64 Bytes cache line
- Non-coherent
- Configuration through mcache_control CSR
- Configurable **modes** :
 - ★ Shared(4KB)
 - ★ Split (2K Each)
 - ★ Scratchpad (3K Scratchpad, 512B Split)
- Vector Processing Unit (**VPU**)
 - 8 lanes, 32 bits per lane
 - 1 FMA unit, 2 IMA unit, 1 INT unit, 1 TRANS unit
 - Extends the RISC-V FP registers from 32b to 256b
- Each FP register is like a vector with **8** elements of **32b** each
- Related **custom extensions** :
 - ★ Packed Single (**PS**)
→ FP32, memory and compute operations
 - ★ Packed Integer (**PI**)
→ Int32/uInt32, compute operations
 - ★ Atomic (**AMO**) → PS and PI AMO operations
 - ★ **Tensor** → Memory, FMA32, FMA16A32, IMA8A32, QUANT, REDUCE operations



Signals

Test Signals

Signal	Direction	Note
ANATEST0	INOUT	Analog Test Pins
ANATEST1	INOUT	Analog Test Pins
TestMode	INPUT	Digital Testmode Enable. For internal use only. Tie to 0 in production.
OSC_CLK_OUT	OUTPUT	Oscillator clk div N

xSPI Target Interface

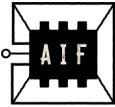
Port	Direction	Note
XS_CS	INPUT	Chip Select
XS_DQ,	INPUT	Data, Bidirectional
XS_RWDS,	INOUT	Data Strobe

UART Target Interface

Port	Direction	Note
Tx	OUTPUT	Transmit Pin
Rx	INPUT	Receive Pin

I2C Controller Interface

Port	Direction	Note
SCL	OUTPUT	Clock
SDA	INOUT	Data. Needs external pullup



QSPI Controller Interface

Port	Direction	Note
CLK	OUTPUT	Clock
DQ[3:0]	INOUT	Data
CS	OUTPUT	Chip select

GPIO Interface.

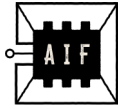
Port	Direction	Note
GPIO[10:0]	INOUT	All GPIO Pins are muxed with other IO protocols.

JTAG Signals

Port	Direction	Note
TCK,	INPUT	Clock. Shared with xSPI
TRSTn	INPUT	Reset. Active Low Shared with xSPI
JTAG_TDI	INPUT	
JTAG_TDO	OUTPUT	
JTAG_TMS	INPUT	

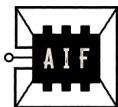
Pin Muxing

A Port	B Port	A_SEL
OSC_CLK_OUT	GPIO[0]	TestMode PIN
SCL	GPIO[1]	system_register.SystemConfig.i2c_enable
SDA	GPIO[2]	system_register.SystemConfig.i2c_enable
SPL_CS	GPIO[3]	system_register.SystemConfig.spi_enable
SPL_CLK	GPIO[4]	system_register.SystemConfig.spi_enable
SPL_DQ[1:0]	GPIO[6:5]	system_register.SystemConfig.spi_enable
SPL_DQ[3:2]	GPIO[8:7]	system_register.SystemConfig.qspi_enable

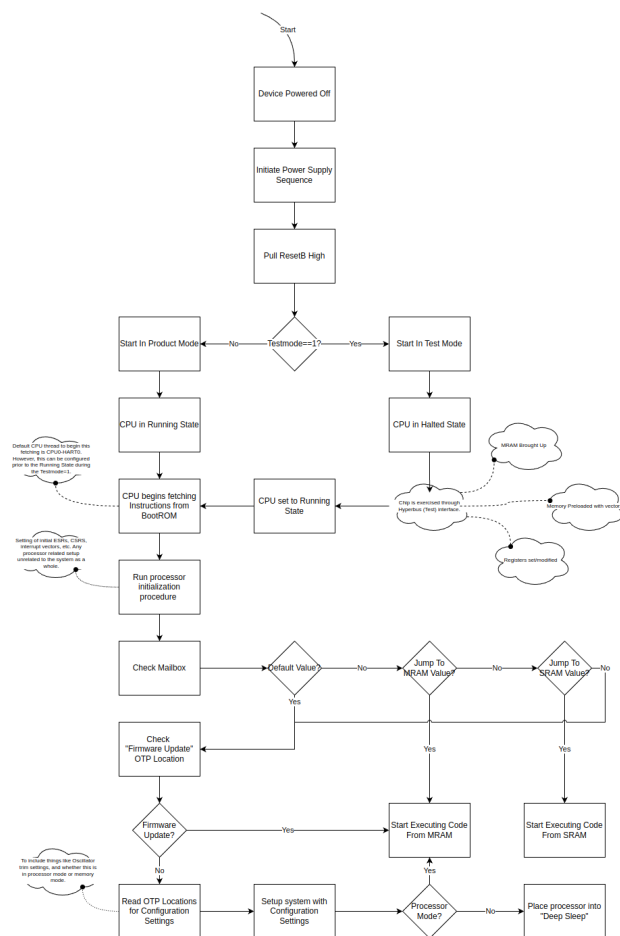


A Port	B Port	A_SEL
UART_TX	GPIO[9]	system_register.SystemConfig.uart_enable
UART_RX	GPIO[10]	system_register.SystemConfig.uart_enable

Note At boot SPI_enable=1 I2C_enable=0, QSPI_enable=0;



Bring up sequence.



Question: How to select single CPU for boot sequence?

Options:

- ✓ - Disabling All threads except for 1
- Powering down CPUs except for 1

How are we saving the values that say which threads are disabled for yield and which thread is being used for bootup?

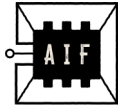
- The yielding/disabled threads are chosen through an MRAM OTP value
- We'll always run CPU0-HART0 during boot up, but this can be overridden if desired during Testmode=1.

In the event of disabled cores, certain instructions that require a fully functioning Neighborhood will also be disabled.

Figure 2: Boot sequence

First boot on the tester.

- TestMode pin is held high. CPU is halted (in reset mode?)
- POR is toggled.
- OSC_CLK.OUT is measured, the PPM difference from expected frequency is calculated.
- Osc TRIM Parameters are calculated. (Index in Trim table?)
- Part is speed graded.
- Osc TRIM parameters and speed grade is written to OTP



Second Boot.

- TestMode is held low.
- CPU Starts executing bootrom code.
 - OTP is read(Trim Param, Speed grade, ROM Update).
 - If ROM Update bit set, jump to updated firmware in MRAM.
 - Trim Ring Oscillator.
 - Program RO-CLK Div for highest clk freq supported by device speed grade.
 - Calculate Clk Div for Periph and MRAM.
 - Program Periph clkdiv and mram clkdiv.
 - Read CHIPMODE.bootloader register to determine location of application code (TCM,MRAM(default))
 - Jump to application code.

Program Mode

- Bring MRAM out of deep sleep.
- Load MRAM/TCM with the application code
- Set CHIPMODE.bootloader register.

Application Code

- If MRAM is in deep sleep mode it needs to be brought out of sleep.
 - Write to `system_registers.PowerDomainReq.mram_dsleep_en` to bring MRAM out of deep sleep.
- Use `system_registers.Mailbox0.mbox0` and `system_registers.Mailbox1.mbox1` for communication from CPU to host chip and vice versa.

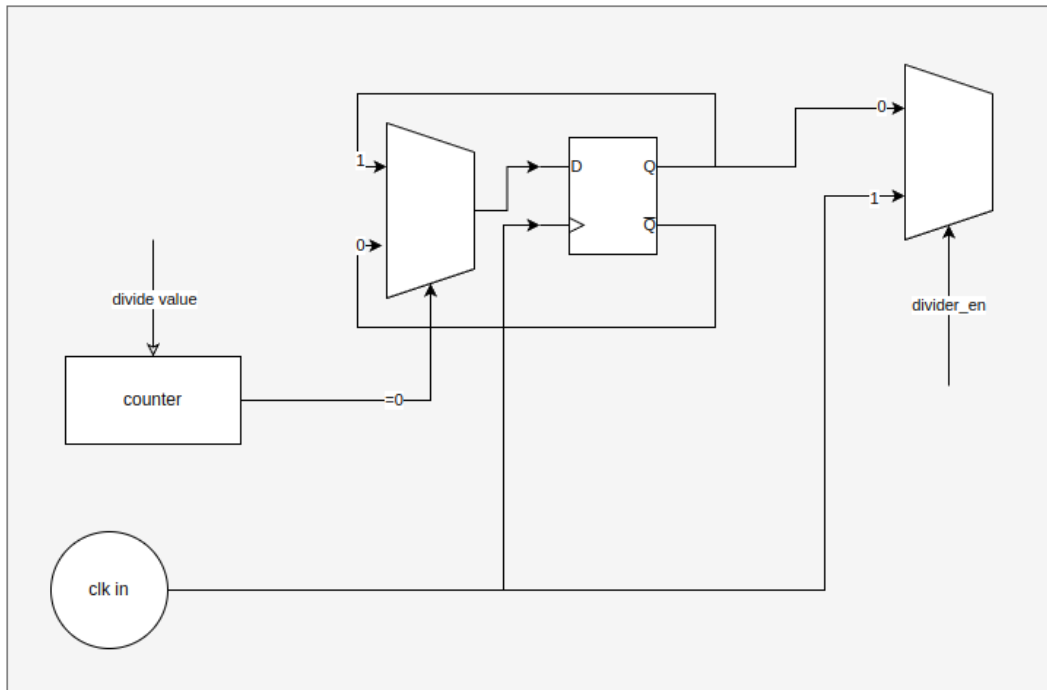
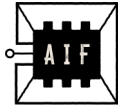
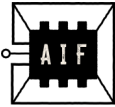


Figure 4: Clock Divider

OSC Clock

- This is a high speed clock generated internally from a ring oscillator.
- The raw ring oscillator(RO) frequency is process dependent and can vary by a few hundred MHz from the designed frequency.
- The first level clock divider is programmed to divide the output of the RO by 32 resulting in a <31 MHz boot clock.
- The tester measures the RO frequency, and process variation and writes it to the OTP. This is used to
 - Trim the RO to generate a 1GHz clock.
 - Program the clock divider to provide the required highspeed CPU Clock.
 - Clock dividers can be reprogrammed later as per the system load.
- A second clock divider divides the **high speed cpu clock** to generate the **system clock**
- Serial protocols require a clock programmed to N times the required baud rate. The Peripheral clock is used as this sampling clock.



xSPI Clock

- This is an external clock with frequency upto 200MHz.
- If required a bit in the xspi register can be written to, to make this the system clock.

CPU Subsystem External Resets

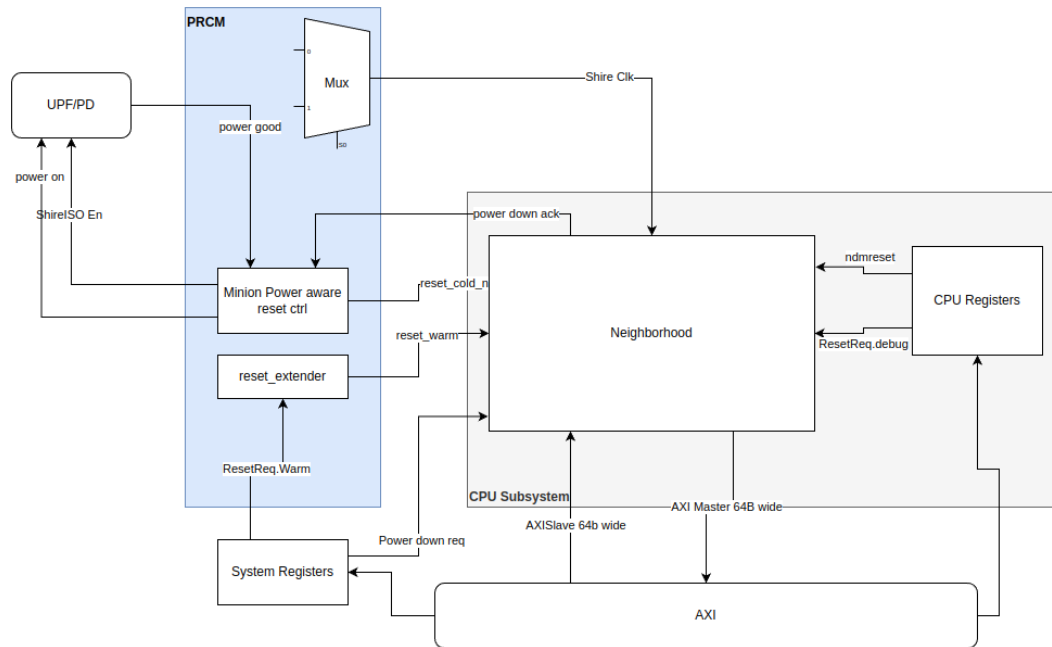


Figure 5: CPU Subsystem Integration

Reset	Description	Register Programming
reset_cold	Full Power on reset	From PRCM (POR,soft_reset, brownout)
reset_warm	Retains state: Does not reset ESR, VC FIFOs etc.	SoftReset.cpu_warm_reset = 0

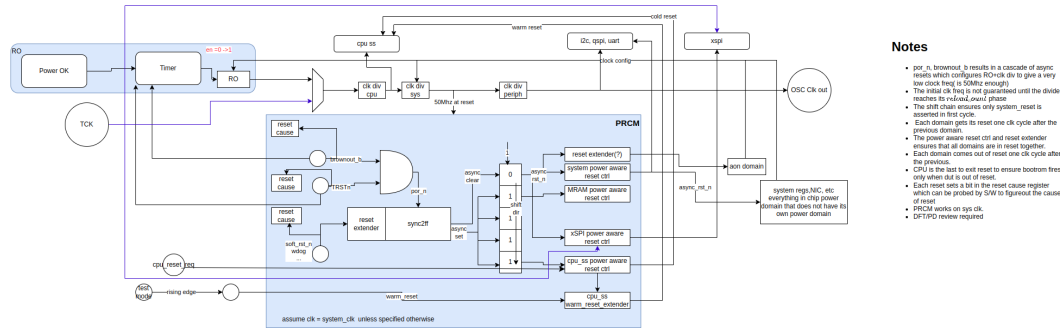
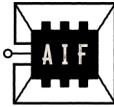


Figure 6: Reset Sequence

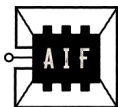
Reset Sequencing

At power on the Ring oscillator and clock dividers have unknown configuration. After reset is applied

- The ring oscillator immediately switched to the default configuration (clock divider enabled, slowest speed).
- Each clock divider latches its configuration(div by 32) only after its internal counter reaches the reload phase.
- The first reset extender ensures that the device remains in reset state until the system divider reloads.
- Power on reset asynchronously resets system registers and xspi ensuring that the correct configuration reaches the ring oscillator, clock dividers and tck/osc_clk mux.
- A Shift register sequences the reset to other modules one by one ensuring that all modules do not enter the reset stage at the same time resulting in a large power drop.
- The reset extender of the power aware reset module ensures that all modules remain in reset for atleast a few common cycles.
- Cpu_system is the last entity to be brought out of reset to ensure that it does not start accessing memory regions that are still in reset.
 - At this point cpu subsystem can start programming the ring oscillator and clock dividers to speed up the clock. ## Power Domains.

The device contains the following power domains.

- Always On.
- System Domain.
- CPU Subsystem Domain.



- Per Minion Power Domain.
- MRAM Digital Domain
- Each Memory (Bootrom, SRAM and MRAM) has its own power domains.
- Each powerdomain is controlled via a power switch.
- At Boot every Digital Domain powerswitch is turned on.
- The boot routine will check the OTP register for domains that need to be powered down after boot and write to the corresponding system_registers.PowerDownReq bits.

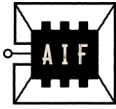
Valid Power State

Simple Power States

State	Sys PS	MRAM Digital PS	MRAM PS	cpu subsys-tem PS	ROM/ SRAM PS	minion PS[7:0]	Notes
All On	On	On	On	On	On	On	
System Off	Off	Off	Off	Off	Off	Off	Extreme low power. Toggle TMS to wake the device
CPU Off	On	X	X	Off	X	Off	
MRAM Off	On	Off	Off	X	X	X	
1-8 Minions Off	On	X	X	On	On	On/Off	
Controlling PowerDomainReq register	system poweroff	mram pd	mram dsleep en	cpu pd	sram pd	minion pd_[0-7]	

Note:

- X: Dont care. The domain is usually on. But can be off if a complex Power State is used.



- Other than the system off state all other power switches can be turned on by clearing the corresponding pd_req bit.
- Complex Power States are:
 - CPU OFF + MRAM OFF
 - Minion_x OFF + MRAM OFF

Power State Controller FSM.

- At boot the FSM is in normal state.
- Writing to pd_req initiates the power down sequence.
- The corresponding domain is informed about the request.
- The domain completes the current outstanding transaction and frees up the system bus and goes into a halt state.
- Domain generates the PD ack signal.
- PD ack signal triggers the Power State Controller which executes the following state changes:
 - ISO_EN State: drives the isolation enable signal to the isolation cells placed on the output wires of the power domain.
 - POWEROFF State: turns off the power switch and remains in this state until pd req goes low, This can be due to:
 - ★ Toggling of TMS signal in case of system power off
 - ★ Clearing the corresponding PD Register bit.
 - When the pd req goes low this state turns on the power switch and transitions to WTGood State.
 - WTGOOD State: It takes some time to ramp up the voltage from 0 to VDD. In this state:
 - ★ A counter is initialized with the system_registers.PowerGood.counter register value and counts down to zero.
 - ★ State changes to ISO_OFF when the counter is zero.
 - ★ soft_reset_req is asserted and stays asserted until we enter the NORMAL state.
 - ISO_OFF: the isolation enable signal to the isolation cells is de-asserted.
 - ★ Note: It is expected the reset values of the domain output signals do not initiate or accept any transaction.
 - RESET: We stay in this state for the number of cycles required to propagate the reset to all sub modules in the power domain.

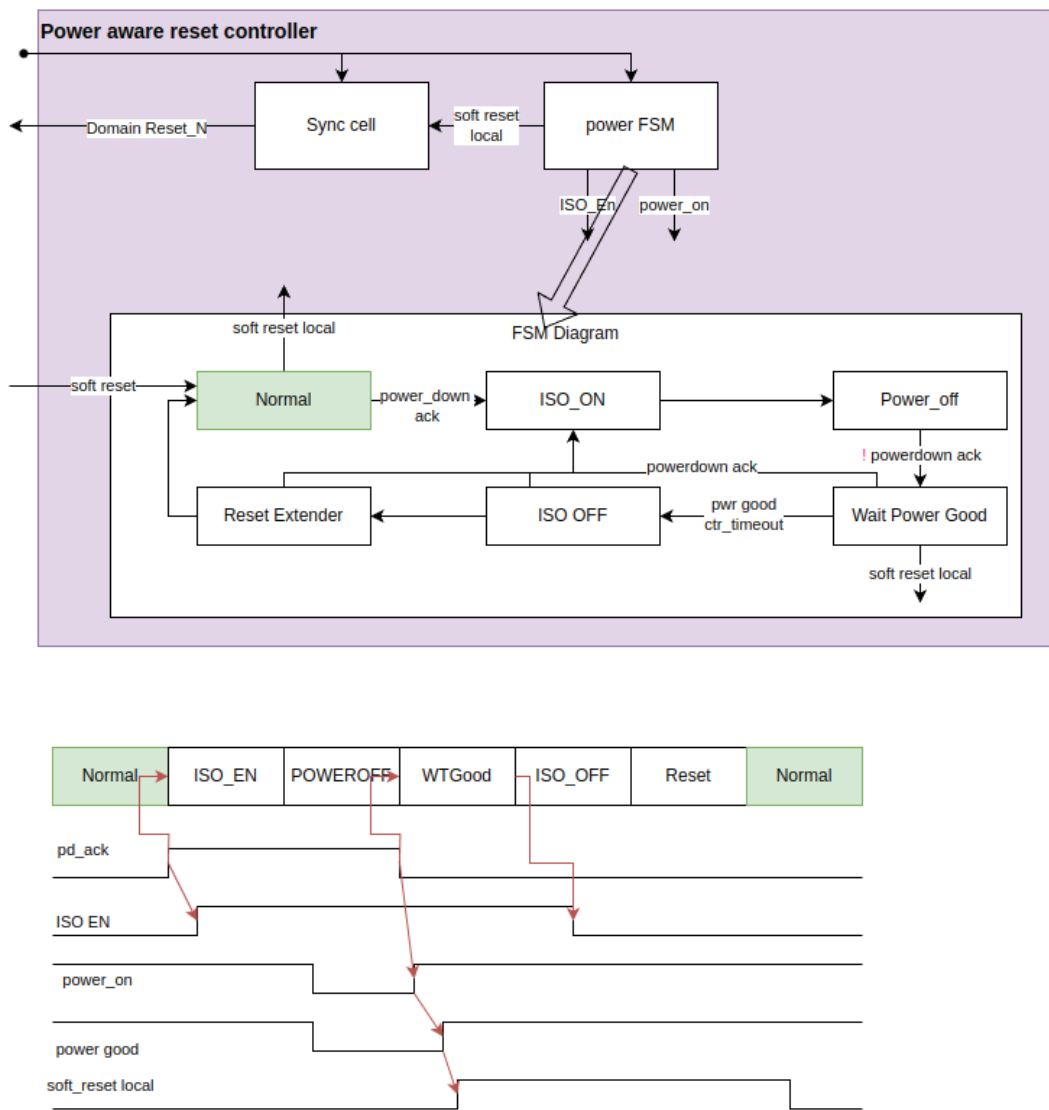
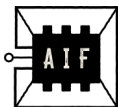
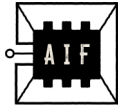
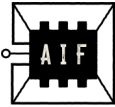


Figure 7: Power Aware reset



Erbium Minion Isotope TRM

- NORMAL: Normal operation resumes.



Interrupts

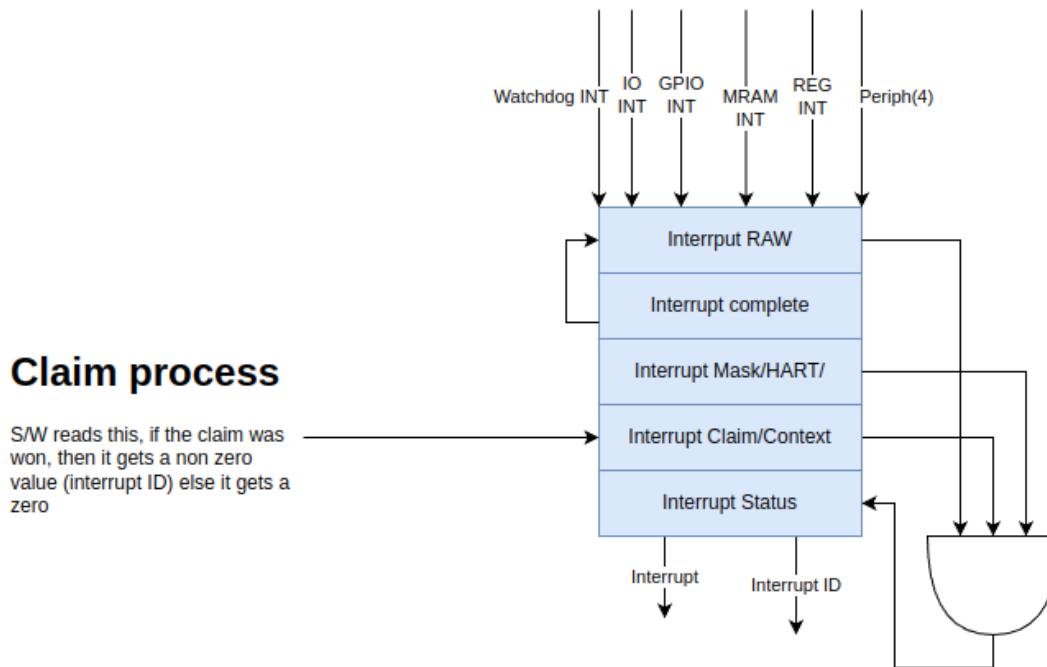


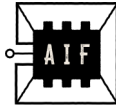
Figure 8: Interrupts

There are two sets of interrupts, Timer and S/W interrupts are described in the minion specification.

This chapter describes the system Interrupts.

The system Interrupts are generated from the following endpoints

- MRAM Interrupt: Raised by MRAM on ECC failure, to trigger a wellness check
- GPIO Interrupt: Raised by the GPIO module when any of the input signal changes state.
- Register Interrupt: A specific bit in the system register that can be written to to trigger an interrupt, e.g. this can be used by xSPI to interrupt the CPU and process the mailbox.
- Periph Interrupt: These interrupts are generated by QSPI, UART, I2C and xSPI Hardware to indicate state changes like TxDone, RxDone, FIFO under/over flow etc.



CPU Handling of system interrupts

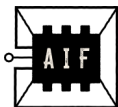
- The system interrupt hardware follow the RISC-V PLIC architecture. The CPU handles the interrupts in Machine mode.
- PLIC MMIO Space consists of the following registers

Register	Count	Purpose
Pending	1bit/interrupt source	Set when source generates an interrupt. input could be level or pulse. Output is level
Mask	1 Reg/HART	The Pending register is Anded with the mask and the result is or'ed and sent to the external interrupt pin of the corresponding core
Claim	1 Reg/HART	A read from this register returns the highest priority interrupt id available for this endpoint.
Complete	1bit/interrupt source	Write of interruptid to this register clears the corresponding interrupt
Priority	2bit/interrupt source	Sets the priority for the interrupt

Interrupt handling process.

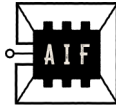
This is a short overview. Refer to the RISC-V privileged ISA for details

- A interrupt from a peripheral sets the corresponding bit in the Pending Register.
- This is ANDed with the per HART mask register and added to the interrupt queue of the HART.
- If the Queue is not empty interrupt signal is sent to to minion.
- If we want to pin down a specific interrupt to a specific minion, we will have to clear its mask bit in all other minions mask registers.
- Multiple interrupts can be outstanding at any given time.
- Each HART that received the interrupt will attempt to read its Claim register. There is one Claim register per HART.
- The PLIC will:
 - Check what is the highest outstanding interrupt in the Queue for this HART and return its ID during the claim read.
 - Remove this interrupt from the Queue of other HART's



CONTENTS

- After servicing the interrupt the minion will write to the Complete register. On Write the PLIC will clear the Pending bit.



Memory Map

Interface	Type
CPU_SUBSYSTEM	AXI
CPU_SUBSYSTEM	APB
xSPI	AXI
xSPI	APB
I2C	AXI4Lite
QSPI	AXI4

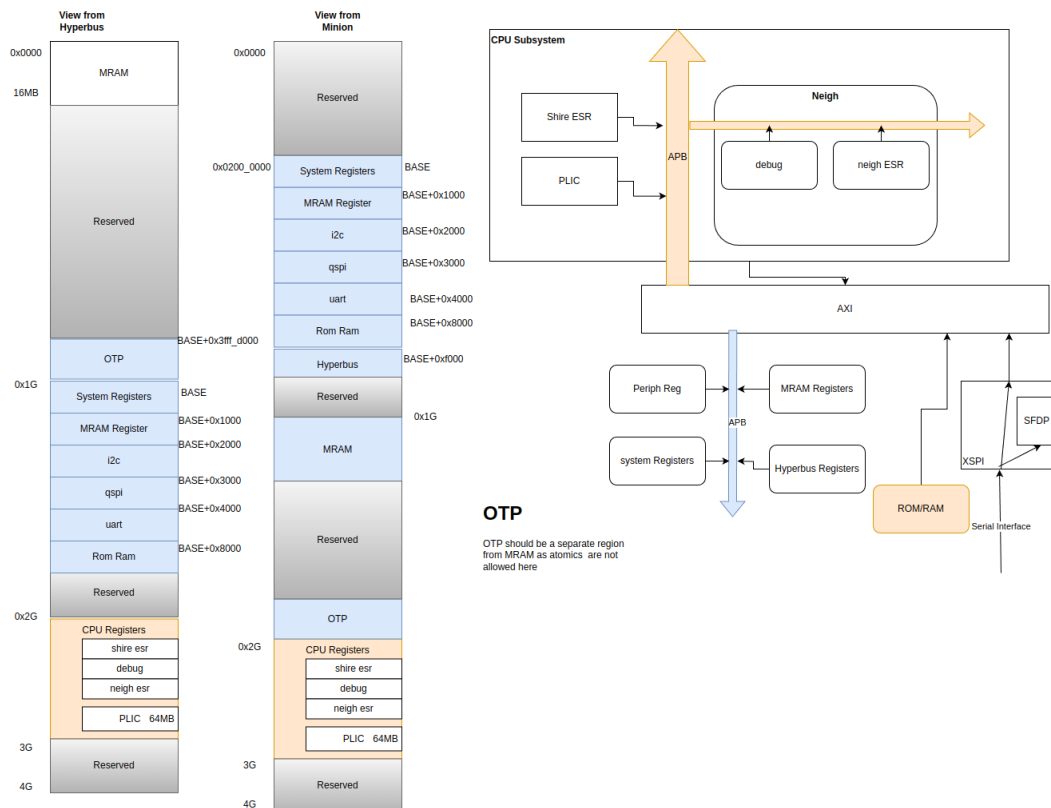


Figure 9: Busmatrix Pathways

Interconnect: Transaction Initiators and Targets

There are 3 Major Transaction Initiators in the System.



1. xSPI
2. CPU
3. UART

Initiator	Interface	Targets Accessible
xSPI	AXI4	MRAM, SRAM, System Registers, Peripheral Registers, HyperBusRegisters, CPURegisters, MRAMRegisters
CPU	AHB-Lite	MRAM, SRAM, Bootrom, System Registers, Peripheral Registers, HyperBusRegisters, CPURegisters, MRAMRegisters
UART	AXI4	MRAM, SystemRegisters

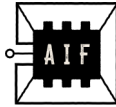
This device can be used as either an Edge AI device or as a simple Flash replacement. The memory Map seen by CPU and UART (Edge AI Mode) is different than the memory map seen by xSPI(Flash replacement device)

CPU and UART Memory Map.

Offset	Identifier	Name
0x02000000	system_registers	System Registers
0x02001000	mrām_registers	Erbium MRAM Block Test Register map.
0x02002000	i2c_registers	—
0x02003000	qspi_registers	—
0x02004000	uart_registers	—
0x0200A000	SRAM	—
0x0200F000	xspi_registers	sccr
0x0E000000	nic_configuration_space	—
0x40000000	mrām	MRAM
0x80000000	cpu_registers	—

xSPI Memory Map.

Customers accessing Erbium through xSPI interface are expected to use it as a MRAM Memory device. In such a case it is natural to assume the memory

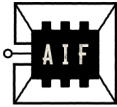


Erbium Minion Isotope TRM

starts from address 0. Hence in Hyperbus case the addressmap is rearranged to facilitate this world view.

Offset	Identifier	Name
0x00000000	mram	MRAM
0x0E000000	nic_configuration_space	—
0x40000000	system_registers	System Registers
0x40001000	mram_registers	Erbium MRAM Block Test Register map.
0x40002000	i2c_registers	—
0x40003000	qspi_registers	—
0x40004000	uart_registers	—
0x40005000	SRAM	—
0x80000000	cpu_registers	—

Note All Register address spaces are 64 bit aligned.



XSPI Datasheet

- xSPI bus interface supporting
 - 1S-1S-1S (SPI)
 - 4S-4D-4D
 - 4D-4D-4D
 - 8D-8D-8D (Octal profile 1)
 - 8D-8D-8D (Hyperbus Octal Profile 2)
- Boot configuration SPI, Quad, Octal Profile 1, Hyperbus.
- Hardware Reset.
- Deep Powerdown mode
- Based on JESD251C, JESD215C-1, JESD216H

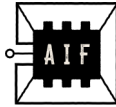
Commands

SPI, Quad SPI, Octal SPI Mode

Commands	Code	1S-1S-1S	4S-4D-4D	8D-8D-8D(Octal)
Read SFDP	5Ah	0.E	1.B	1.B
Read (0L)	NA	NA	NA	NA
Read Register	65	0.F	1.B	1.B
Write Register	71	0.K	1.D	1.D
Reset Device	99h	0.A	1.A	1.A
and Enter default mode(1S)				
Read Memory	0Bh	0.F	1.B	1.B
Write Memory	02h	0.K	1.D	1.D
setRate	52h	0.G	1.A	1.A

SetRate command writes a tuple of 3 bytes, each byte specifies the rate for cmd, addr, data. The encoding is

- S1=0,
- D1=1,
- S2=not Implemented,
- D2 =NotImplemented,
- S4=4,
- D4=5,



- S8=6,
- D8=7,
- HB=8

Format 1S-1S-1S

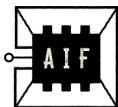
The following formats are applicable to 1S-1S-1S mode of operation.

Format	Ext	Address Bytes	Latency	Read Txn	Write Txn	Note
0.A	✗	✗	✗	✗	✗	CMD Only
0.E	✗	3	✓	✓	✗	Read Reg
0.F	✗	4	✓	✓	✗	Read Mem
0.G	✗	✗	✗	✗	✓	setRate
0.J	✗	3	✗	✗	✓	Write Reg
0.K	✗	4	✗	✗	✓	Write Mem

Format Other rates

- The following formats are applicable to 4S/4D/8S/8D modes of operation.
- These modes require Ext and address is always 4.

Format	Ext	Address Bytes	Latency	Read Txn	Write Txn	Note
1.A	✓	✗	✗	✗	✗	CMD Only
1.B	✓	4	✓	✓	✗	Read Reg/Mem
1.D	✓	4	✗	✗	✓	Write Reg/Mem
1.G	✓	✗	✗	✗	✓	setRate



Default Mode.

The standard allows the datarate at reset to be something other than SPI(1S-1S-1S). This can be controlled via the `default_mode` pin.

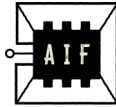
Default Mode	Default Rate
00	Hyperbus
01	Octal SPI
10	Quad SPI
11	SPI

For Hyperbus mode all 3 data bytes of setrate should be HB.

- SPI Memory access uses 4B addressing.
- SPI Register access uses 3B addressing.
- Some commands marked 1.A do not require an address field. e.g. Enter Pd, Exit PD, Reset Device.

Note

- 1.G is a non standard format for set rate
- All writes have zero latency
- All Reads have latency cycles.
- 1s can have 3 or 4 byte address, use 3 byte for xspi-register and 4 byte for memory transactions
- non 1S only support 4 byte address.
- The 1.X commands follow the pattern of CMD-Ext, 4B Address, X cycles of latency, Data
- The 0.X commands do not have Ext
- For Register Transaction the Data is 32 bit.
- For Memory Transaction Data is multiple of 64 bit.
- Behavior when data width is not multiple of 64 (for mem) or 32 for reg is not defined.



- We will not implement read zero latency as that will not work with our pipeline.
- Supported(Tested) Modes are (ref 6.10.18 of SFDP)
 - * 1S-1S-1S
 - * 8D-8D-8D
 - * 8S-8S-8S
 - * 4S-4D-4D
 - * 4D-4D-4D
 - * 4S-4S-4S

Not supported in Phase 1

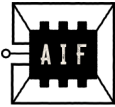
Mode	Plan
Wrapped bursts	(Phase 2)
XIP	(Phase 2)
Write Enable	Not Supported
Page/Sector Erase/Program	Not Supported

Verification Requirement.

- Verify all supported mode.
- For spi, quadspi, octalspi do a cross coverage of command, rates, latency, burst length, mem/reg, read/write
- Verify SFPD.
- Verify contents and behavior of register space.
- Ensure register space is accessible from both minion and xspi.
- verify hyperbus ensure cross coverage for latency and burst
- Ensure all functionality is implemented.

Booting.

- The device requires a few cycles of xspi clock to complete the reset process.
- Whenever the XSPI IP is resetted e.g. por, soft_reset etc. Run TCK for 8 cycles.
- If the xspi controller is not capable of running TCK without a transaction, issue a transaction with cmd=0



Error handling.

Errors are generated in cases where

1. Host tries to access invalid memory space.
2. Host transfer does not complete due to network congestion.

`sccr.interrupt_status` register should be checked after each transaction for errors. The fields are as follows.

- `axi_resp`: The last known Error from the interconnect.
- `axi_read_underflow`: Set when a read transaction is issued and read data is not available before end of latency period or the next word of a burst is not available at the time when the last bit of current burst is sent on wire.
- `axi_write_overflow`: Set when the host issues a write burst transaction and the next word is presented before the current word is consumed by the interconnect.

Once set these bits remain set and can be cleared by reading the interrupt register.

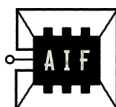
If the `sccr.xspi_control.interrupt_enable` bit is set, a non zero value in the `interrupt_status` register can generate an interrupt to the minion. It is recommended to keep the interrupt disabled and let the host manage the exception. Only in case where the host cannot manage exceptions should minion step in.

Recommended host flow.

1. For write operations, Poll `sccr.xsip_status.wip` until the bit is cleared.
2. Check the value of `sccr.interrupt_status` if it is non zero initiate error recovery (e.g. retry the transaction at a lower data rate, increase latency, reduce burst size, reduce clock frequency, check the destination address etc.)
3. If it is zero, write to the mailbox to inform the minion the transaction is over and it can take control of the buffer.

Errata:

There are errors in the JEDC 251C Specification.

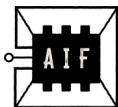


- The Profile2 Hyperbus bit map assigns A31:A3 to a 24 bit field instead of a 29 bit field. The upper 5 bits are marked as reserved. We follow the Cypress Hyperbus bitfield here and use the upper 5 bits to form the required 29 bit address address.
- In 251C-1 the Specification for QuadSPI defines a format for Quad SPI but does not use it. We are ignoring the format and using 1.X format for Quad mode.

SFDP_Reg address map

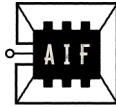
- Absolute Address: 0x0
- Base Offset: 0x0
- Size: 0xB3C

Offset	Identifier	Name
0x000	reg6_2_1	—
0x004	reg6_2_2	—
0x008	reg6_4_1	—
0x00C	reg6_4_2	—
0x010	reg6_7_1	—
0x014	reg6_7_2	—
0x018	reg6_9_1	—
0x01C	reg6_9_2	—
0x020	reg6_10_1	—
0x024	reg6_10_2	—
0x028	reg6_11_1	—
0x02C	reg6_11_2	—
0x030	reg6_17_1	—
0x034	reg6_17_2	—
0x400	reg_6_4_4	—
0x404	reg_6_4_5	—
0x408	reg6_4_6	—
0x40C	reg6_4_7	—
0x410	reg6_4_8	—
0x414	reg6_4_9	—
0x418	reg6_4_10	—
0x41C	reg6_4_11	—
0x420	reg6_4_12	—



CONTENTS

Offset	Identifier	Name
0x424	reg6_4_13	—
0x428	reg6_4_14	—
0x42C	reg6_4_15	—
0x430	reg6_4_16	—
0x434	reg6_4_17	—
0x438	reg6_4_18	—
0x43C	reg_6_4_19	—
0x440	reg_6_4_20	—
0x444	reg_6_4_21	—
0x448	reg_6_4_22	—
0x44C	reg_6_4_23	—
0x450	reg_6_4_24	—
0x454	reg_6_4_25	—
0x458	reg_6_4_26	—
0x700	reg_6_7_3	—
0x704	reg_6_7_4	—
0x900	reg_6_9_3	—
0x904	reg_6_9_4	—
0x908	reg_6_9_5	—
0xA00	reg_6_10_3	—
0xA04	reg_6_10_4	—
0xA08	reg_6_10_5	—
0xA0C	reg_6_10_6	—
0xA10	reg_6_10_7	—
0xA14	reg_6_10_8	—
0xA18	reg_6_10_9	—
0xA1C	reg_6_10_10	—
0xA20	reg_6_10_11	—
0xA24	reg_6_10_12	—
0xA28	reg_6_10_13	—
0xA2C	reg_6_10_14	—
0xA30	reg_6_10_15	—
0xA34	reg_6_10_16	—
0xA38	reg_6_10_17	—
0xA3C	reg_6_10_18	—
0xA40	reg_6_10_19	—
0xA44	reg_6_10_20	—
0xA48	reg_6_10_21	—



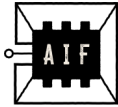
Offset	Identifier	Name
0xA4C	reg_6_10_22	—
0xA50	reg_6_10_23	—
0xA54	reg_6_10_24	—
0xA58	reg_6_10_25	—
0xA5C	reg_6_10_26	—
0xA60	reg_6_10_27	—
0xA64	reg_6_10_28	—
0xA68	reg_6_10_29	—
0xA6C	reg_6_11_3	—
0xA70	reg_6_11_4	—
0xB00	reg_6_11_5	—
0xB04	reg_6_11_6	—
0xB08	reg_6_11_7	—
0xB0C	reg_6_11_8	—
0xB10	reg_6_11_9	—
0xB14	reg_6_11_10	—
0xB18	reg_6_11_11	—
0xB1C	reg_6_11_12	—
0xB20	reg_6_11_13	—
0xB24	reg_6_11_14	—
0xB28	reg_6_11_15	—
0xB2C	reg_6_17_3	—
0xB30	reg_6_17_4	—
0xB34	reg_6_17_5	—
0xB38	reg_6_17_6	—

reg6_2_1 register

- Absolute Address: 0x0
- Base Offset: 0x0
- Size: 0x4

Bits	Identifier	Access	Reset	Name
31:0	signature	r	0x50444653	—

signature field SFDP Signature

**reg6_2_2 register**

- Absolute Address: 0x4
- Base Offset: 0x4
- Size: 0x4

Bits	Identifier	Access	Reset	Name
7:0	minor	r	0xC	—
15:8	major	r	0x1	—
23:16	numHdr	r	0x6	—
31:24	access_protocol	r	0xFA	—

minor field SFDP Version Minor

major field SFDP Version Major

numHdr field Num Parameter Header TBD

access_protocol field TBD SFDP Access protocol

reg6_4__1 register

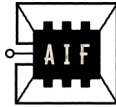
- Absolute Address: 0x8
- Base Offset: 0x8
- Size: 0x4

Bits	Identifier	Access	Reset	Name
7:0	id	r	0x0	—
15:8	minor_rev	r	0x9	—
23:16	major_rev	r	0x1	—
31:24	tbl_len	r	0x17	—

id field parameter id lsb

minor_rev field minor rev

major_rev field major_rev



tbl_len field Tbl Len

reg6_4__2 register

- Absolute Address: 0xC
- Base Offset: 0xC
- Size: 0x4

Bits	Identifier	Access	Reset	Name
23:0	param_tbl_ptr	r	0x400	—
31:24	param_id_msb	r	0xFF	—

param_tbl_ptr field Parameter Table Pointer TBD

param_id_msb field Parameter Table ID MSB

reg6_7__1 register

- Absolute Address: 0x10
- Base Offset: 0x10
- Size: 0x4

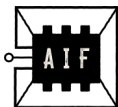
Bits	Identifier	Access	Reset	Name
7:0	id	r	0x84	—
15:8	minor_rev	r	0x1	—
23:16	major_rev	r	0x1	—
31:24	tbl_len	r	0x2	—

id field parameter id lsb

minor_rev field minor rev

major_rev field major_rev

tbl_len field Tbl Len

**reg6_7__2 register**

- Absolute Address: 0x14
- Base Offset: 0x14
- Size: 0x4

Bits	Identifier	Access	Reset	Name
23:0	param_tbl_ptr	r	0x700	—
31:24	param_id_msb	r	0xFF	—

param_tbl_ptr field Parameter Table Pointer TBD

param_id_msb field Parameter Table ID MSB

reg6_9__1 register

- Absolute Address: 0x18
- Base Offset: 0x18
- Size: 0x4

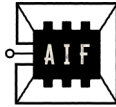
Bits	Identifier	Access	Reset	Name
7:0	id	r	0x6	—
15:8	minor_rev	r	0x0	—
23:16	major_rev	r	0x1	—
31:24	tbl_len	r	0x3	—

id field parameter id lsb

minor_rev field minor rev

major_rev field major_rev

tbl_len field Tbl Len

**reg6_9__2 register**

- Absolute Address: 0x1C
- Base Offset: 0x1C
- Size: 0x4

Bits	Identifier	Access	Reset	Name
23:0	param_tbl_ptr	r	0x900	—
31:24	param_id_msb	r	0xFF	—

param_tbl_ptr field Parameter Table Pointer TBD

param_id_msb field Parameter Table ID MSB

reg6_10__1 register

- Absolute Address: 0x20
- Base Offset: 0x20
- Size: 0x4

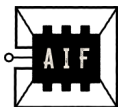
Bits	Identifier	Access	Reset	Name
7:0	id	r	0x87	—
15:8	minor_rev	r	0x1	—
23:16	major_rev	r	0x1	—
31:24	tbl_len	r	0x1C	—

id field parameter id lsb

minor_rev field minor rev

major_rev field major_rev

tbl_len field Tbl Len

**reg6_10__2 register**

- Absolute Address: 0x24
- Base Offset: 0x24
- Size: 0x4

Bits	Identifier	Access	Reset	Name
23:0	param_tbl_ptr	r	0xA00	—
31:24	param_id_msb	r	0xFF	—

param_tbl_ptr field Parameter Table Pointer TBD

param_id_msb field Parameter Table ID MSB

reg6_11__1 register

- Absolute Address: 0x28
- Base Offset: 0x28
- Size: 0x4

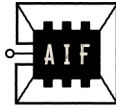
Bits	Identifier	Access	Reset	Name
7:0	id	r	0x9	—
15:8	minor_rev	r	0x0	—
23:16	major_rev	r	0x1	—
31:24	tbl_len	r	0xD	—

id field parameter id lsb

minor_rev field minor rev

major_rev field major_rev

tbl_len field Tbl Len

**reg6_11__2 register**

- Absolute Address: 0x2C
- Base Offset: 0x2C
- Size: 0x4

Bits	Identifier	Access	Reset	Name
23:0	param_tbl_ptr	r	0xB00	—
31:24	param_id_msb	r	0xFF	—

param_tbl_ptr field Parameter Table Pointer TBD

param_id_msb field Parameter Table ID MSB

reg6_17__1 register

- Absolute Address: 0x30
- Base Offset: 0x30
- Size: 0x4

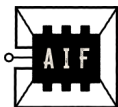
Bits	Identifier	Access	Reset	Name
7:0	id	r	0xF	—
15:8	minor_rev	r	0x1	—
23:16	major_rev	r	0x1	—
31:24	tbl_len	r	0xA	—

id field parameter id lsb

minor_rev field minor rev

major_rev field major_rev

tbl_len field Tbl Len

**reg6_17_2 register**

- Absolute Address: 0x34
- Base Offset: 0x34
- Size: 0x4

Bits	Identifier	Access	Reset	Name
23:0	param_tbl_ptr	r	0x1100	—
31:24	param_id_msb	r	0xFF	—

param_tbl_ptr field Parameter Table Pointer TBD

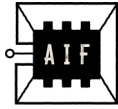
param_id_msb field Parameter Table ID MSB

reg_6_4_4 register

- Absolute Address: 0x400
- Base Offset: 0x400
- Size: 0x4

Bits	Identifier	Access	Reset	Name
1:0	erase_size	r	0x3	—
2	write_granularity	r	0x0	—
3	always_volatile_csr	r	0x1	—
4	we_instruction	r	0x1	—
7:5	unused1	r	0x7	—
15:8	erase_4kb	r	0xFF	—
16	fs_1s1s2s	r	0x0	—
18:17	addrBytes	r	0x2	—
19	dtr_mode	r	0x1	—
20	fs_1s2s2s	r	0x0	—
21	fs_1s4s4s	r	0x0	—
22	fs_1s1s1s	r	0x0	—
31:23	unused0	r	0xFF	—

erase_size field Block/Sector Erase size.



write_granularity field Write Granularity (1byte).

always_volatile_csr field Volatile Status Register Block Protect.

we_instruction field Write Enable Instruction select for writing to volatile status reg.

unused1 field Unused

erase_4kb field Does not support 4KB Erase

fs_1s1s2s field Does not support 1S-1S-2S Fast Read

addrBytes field Address Bytes (4-Byte Only Addressing)

dtr_mode field Supports DTR mode

fs_1s2s2s field Does not support 1S-2S-2S Fast Read

fs_1s4s4s field Does not support 1S-4S-4S Fast Read

fs_1s1s1s field Does not support 1S-1S-4S Fast Read

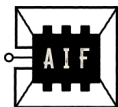
unused0 field Unused

reg_6_4_5 register

- Absolute Address: 0x404
- Base Offset: 0x404
- Size: 0x4

Bits Identifier	Access Reset	Name
31:0 mem_density r	0xFFFFFFFF	—

mem_density field memory Density(in bits)

**reg6_4_6 register**

- Absolute Address: 0x408
- Base Offset: 0x408
- Size: 0x4

Bits	Identifier	Access	Reset	Name
4:0	waitstate_1s4s4s	r	0x2	—
7:5	mode_1s4s4s	r	0x0	—
15:8	fr_inst_1s4s4s	r	0xAB	—
20:16	waitstate_1s1s4s	r	0x2	—
23:21	mode_1s1s4s	r	0x0	—
31:24	fr_inst_1s1s4s	r	0xAA	—

waitstate_1s4s4s field 1s-4s-4s fast read wait states

mode_1s4s4s field 1s-4s-4s fast read num mode clocks

fr_inst_1s4s4s field 1s-4s-4s fast read wait states

waitstate_1s1s4s field 1s-1s-4s fast read wait states

mode_1s1s4s field 1s-1s-4s fast read num mode clocks

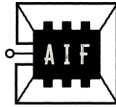
fr_inst_1s1s4s field 1s-1s-4s fast read instruction

reg6_4_7 register

- Absolute Address: 0x40C
- Base Offset: 0x40C
- Size: 0x4

Bits	Identifier	Access	Reset	Name
31:0	reserved	r	0x0	—

reserved field 2S mode is not supported. Reserved

**reg6_4_8 register**

- Absolute Address: 0x410
- Base Offset: 0x410
- Size: 0x4

Bits Identifier Access Reset Name				
0	mode_2s	r	0x0	—
4	mode_4s	r	0x1	—

mode_2s field Supports 2S-2S-2S fast read mode(not supported)

mode_4s field Supports 4S-4S-4S fast read mode

reg6_4_9 register

- Absolute Address: 0x414
- Base Offset: 0x414
- Size: 0x4

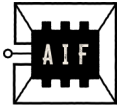
Bits Identifier Access Reset Name				
31:0	reserved	r	0x0	—

reserved field 2S mode is not supported. Reserved

reg6_4_10 register

- Absolute Address: 0x418
- Base Offset: 0x418
- Size: 0x4

Bits	Identifier	Access	Reset	Name
20:16	waitstate_4s4s4s	r	0x2	—
23:21	mode_4s4s4s	r	0x0	—
31:24	fr_inst_4s4s4s	r	0xAC	—



waitstate_4s4s4s field 4s-4s-4s fast read wait states

mode_4s4s4s field 4s-4s-4s fast read num mode clocks

fr_inst_4s4s4s field 4s-4s-4s fast read instruction

reg6_4_11 register

- Absolute Address: 0x41C
- Base Offset: 0x41C
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

reserved field Erase instruction 2,1 is not supported. Reserved

reg6_4_12 register

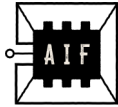
- Absolute Address: 0x420
- Base Offset: 0x420
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

reserved field Erase instruction 4,3 is not supported. Reserved

reg6_4_13 register

- Absolute Address: 0x424
- Base Offset: 0x424
- Size: 0x4



Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

reserved field Erase instruction is not supported. Reserved

reg6_4_14 register

- Absolute Address: 0x428
- Base Offset: 0x428
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

reserved field Erase/Program instruction is not supported. Reserved

reg6_4_15 register

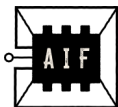
- Absolute Address: 0x42C
- Base Offset: 0x42C
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

reserved field Suspend/Resume for Erase/Program instruction is not supported. Reserved

reg6_4_16 register

- Absolute Address: 0x430
- Base Offset: 0x430
- Size: 0x4



Bits	Identifier	Access	Reset	Name
31:0	reserved	r	0x0	—

reserved field Suspend/Resume for Erase/Program instruction is not supported. Reserved

reg6_4_17 register

- Absolute Address: 0x434
- Base Offset: 0x434
- Size: 0x4

Bits	Identifier	Access	Reset	Name
1:0	reserved	r	0x0	—
7:2	dev_busy_poll	r	0x0	—
12:8	exit_delay_count	r	0x2	—
14:13	exit_delay_units	r	0x3	—
22:15	inst_deep_power_down_exit	r	0xAE	—
30:23	inst_deep_power_down_enter	r	0xAD	—
31	deep_power_down	r	0x1	—

reserved field Reserved

dev_busy_poll field device busy poll (not supported)

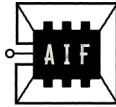
exit_delay_count field Exit Deep powerdown delay count

exit_delay_units field Exit Deep powerdown delay units 64us

inst_deep_power_down_exit field Exit Deep powerdown Instruction

inst_deep_power_down_enter field Enter Deep powerdown Instruction

deep_power_down field Deep powerdown is supported

**reg6_4_18 register**

- Absolute Address: 0x438
- Base Offset: 0x438
- Size: 0x4

Bits	Identifier	Access	Reset	Name
3:0	mode_disable444	r	0x0	—
8:4	mode_enable444	r	0x0	—
9	xip_supported_044	r	0x1	—
15:10	mode_exit044	r	0x0	—
19:16	mode_entry044	r	0x0	—
22:20	quad_enable	r	0x0	—
23	hold_rst_support	r	0x0	—

mode_disable444 field 4-4-4 Mode disable sequence TBD

mode_enable444 field 4-4-4 Mode enable sequence TBD

xip_supported_044 field 0-4-4 Mode supported (XIP)

mode_exit044 field 0-4-4 Mode exit method TBD

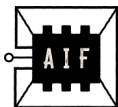
mode_entry044 field 0-4-4 Mode entry method TBD

quad_enable field quad enable requirement

hold_rst_support field hold/reset disable (not supported)

reg_6_4_19 register

- Absolute Address: 0x43C
- Base Offset: 0x43C
- Size: 0x4



Bits	Identifier	Access	Reset	Name
6:0	volatile_status_reg_1	r	0x2	—
13:8	soft_reset_support	r	0x0	—
23:14	exit_4B_addressing	r	0x0	—
31:24	enter_4B_addressing	r	0x40	—

volatile_status_reg_1 field volatile register status1

soft_reset_support field Soft reset instruction support (not supported)

exit_4B_addressing field Exit 4 byte addressing TBD

enter_4B_addressing field Enter 4 byte addressing; Always in \$B mode

reg_6_4_20 register

- Absolute Address: 0x440
- Base Offset: 0x440
- Size: 0x4

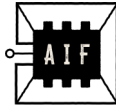
Bits	Identifier	Access	Reset	Name
4:0	waitstate_1s8s8s	r	0x2	—
7:5	mode_1s8s8s	r	0x0	—
15:8	fr_inst_1s8s8s	r	0xAB	—
20:16	waitstate_1s1s8s	r	0x2	—
23:21	mode_1s1s8s	r	0x0	—
31:24	fr_inst_1s1s8s	r	0xAF	—

waitstate_1s8s8s field 1s-8s-8s fast read wait states

mode_1s8s8s field 1s-8s-8s fast read num mode clocks

fr_inst_1s8s8s field 1s-8s-8s fast read wait states

waitstate_1s1s8s field 1s-1s-8s fast read wait states



mode_1s1s8s field 1s-1s-8s fast read num mode clocks

fr_inst_1s1s8s field 1s-1s-8s fast read instruction

reg_6_4_21 register

- Absolute Address: 0x444
- Base Offset: 0x444
- Size: 0x4

Bits	Identifier	Access	Reset	Name
22:18	drive_strength	r	0x1	—
23	jedec_reset	r	0x0	—
25:24	data_strobe_str_waveform	r	0x0	—
26	data_strobe_support_4s4S4S	r	0x0	—
27	data_strobe_support_4s4d4d	r	0x0	—
30:29	cmd_ext	r	0x0	—
31	byte_order_8D	r	0x0	—

drive_strength field Drive Strength TBD

jedec_reset field JEDEC reset(not supported)

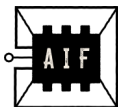
data_strobe_str_waveform field Data strobe STR Waveform TBD

data_strobe_support_4s4S4S field Data strobe support in QPI 4S-4S-4S

data_strobe_support_4s4d4d field Data strobe support in QPI 4S-4D-4D

cmd_ext field Octal DTR Command/Extension

byte_order_8D field Byte order in 8D-8D-8D mode same as 1S-1S-1S mode

**reg_6_4_22 register**

- Absolute Address: 0x448
- Base Offset: 0x448
- Size: 0x4

Bits	Identifier	Access	Reset	Name
3:0	disable_seq_8s8s8s	r	0x1	—
8:4	enable_seq_8s8s8s	r	0x2	—
9	xip_supported_088	r	0x1	—
15:10	xip_exit_088	r	0x4	—
19:16	xip_entry_088	r	0x1	—
22:20	octal_enable_req	r	0x1	—

disable_seq_8s8s8s field 8s8s8s mode disable seq

enable_seq_8s8s8s field 8s8s8s mode enable seq

xip_supported_088 field 0-8-8 XIP supported

xip_exit_088 field 0-8-8 Mode Exit method TBD

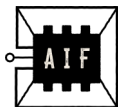
xip_entry_088 field 0-8-8 Mode Entry method

octal_enable_req field octal.enable req

reg_6_4_23 register

- Absolute Address: 0x44C
- Base Offset: 0x44C
- Size: 0x4

Bits	Identifier	Access	Reset	Name
3:0	max_4S_speed_without_ds	r	0x8	—
7:4	max_4S_speed_with_ds	r	0x7	—
11:8	max_4D_speed_without_ds	r	0x8	—
15:12	max_4D_speed_with_ds	r	0x8	—



Bits	Identifier	Access	Reset	Name
19:16	max_8S_speed_without_ds	r	0x8	—
23:20	max_8S_speed_with_ds	r	0x8	—
27:24	max_8D_speed_without_ds	r	0x8	—
31:28	max_8D_speed_with_ds	r	0x8	—

max_4S_speed_without_ds field max 4S-4S-4S speed not using DS 200 Mhz

max_4S_speed_with_ds field max 4S-4S-4S speed using DS 166Mhz

max_4D_speed_without_ds field max 4S-4D-4D speed not using DS

max_4D_speed_with_ds field max 4S-4D-4D speed using DS

max_8S_speed_without_ds field max 8S speed not using DS

max_8S_speed_with_ds field max 8S speed using DS 200Mhz

max_8D_speed_without_ds field max 8D speed not using DS

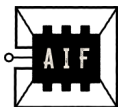
max_8D_speed_with_ds field max 8D speed using DS 200Mhz

reg_6_4_24 register

- Absolute Address: 0x450
- Base Offset: 0x450
- Size: 0x4

Bits	Identifier	Access	Reset	Name
2:0	supports_4s4d4d	r	0x1	—
4:3	supports_1s4d4d	r	0x1	—
5	supports_1s2d2d	r	0x0	—
6	supports_1s1d1d	r	0x0	—

supports_4s4d4d field supports 4S-4D-4D Fast Read



supports_1s4d4d field supports 1S-4D-4D Fast Read

supports_1s2d2d field supports 1S-2D-2D Fast Read

supports_1s1d1d field supports 1S-1D-1D Fast Read

reg_6_4_25 register

- Absolute Address: 0x454
- Base Offset: 0x454
- Size: 0x4

Bits	Identifier	Access	Reset	Name
31:0	reserved	r	0x0	—

reserved field 1S2D2D and 1S1D1D not supported

reg_6_4_26 register

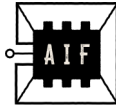
- Absolute Address: 0x458
- Base Offset: 0x458
- Size: 0x4

Bits	Identifier	Access	Reset	Name
4:0	waitstate_1s4d4d	r	0x2	—
7:5	mode_1s4d4d	r	0x0	—
15:8	fr_inst_1s4d4d	r	0xB1	—
20:16	waitstate_4s4d4d	r	0x2	—
23:21	mode_4s4d4d	r	0x0	—
31:24	fr_inst_4s4d4d	r	0xB0	—

waitstate_1s4d4d field 1s-4d-4d fast read wait states

mode_1s4d4d field 1s-4d-4d fast read num mode clocks

fr_inst_1s4d4d field 1s-4d-4d fast read wait states



waitstate_4s4d4d field 4s-4d-4d fast read wait states

mode_4s4d4d field 4s-4d-4d fast read num mode clocks

fr_inst_4s4d4d field 4s-4d-4d fast read instruction

reg_6_7_3 register

- Absolute Address: 0x700
- Base Offset: 0x700
- Size: 0x4

Bits	Identifier	Access	Reset	Name
0	r_inst_1s1s1s	r	0x1	—
1	fr_inst_1s1s1s	r	0x1	—
3:2	reserved_2s	r	0x0	—
4	fr_inst_1s1s4s	r	0x1	—
5	fr_inst_1s4s4s	r	0x1	—
12:6	erase_reserved	r	0x0	—
13	fr_inst_1s1d1d	r	0x0	—
14	fr_inst_1s2d2d	r	0x0	—
15	fr_inst_1s4d4d	r	0x0	—
19:16	sector_reserved	r	0x0	—
20	fr_inst_1s1s8s	r	0x0	—
21	fr_inst_1s8s8s	r	0x0	—
22	fr_inst_1s8d8d	r	0x0	—
23	page_program_support_1s1s8s	r	0x0	—
24	page_program_support_1s8s8s	r	0x0	—

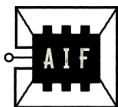
r_inst_1s1s1s field 1s1s1s Read cmd inst 13h support(yes)

fr_inst_1s1s1s field 1s1s1s Fast Read cmd inst 0Ch support(yes)

reserved_2s field 2S not supported

fr_inst_1s1s4s field 1s1s4s fast Read cmd inst 34h support(yes)

fr_inst_1s4s4s field 1s4s4s fast Read cmd inst 3Eh support(yes)



erase_reserved field Erase/Page cmds not supported

fr_inst_1s1d1d field 1s1d1d DTR Read cmd inst 0E support(no)

fr_inst_1s2d2d field 1s2d2d DTR Read cmd inst BEh support(no)

fr_inst_1s4d4d field 1s4d4d DTR Read cmd inst EEh support(no)

sector_reserved field Sector cmds not supported

fr_inst_1s1s8s field 1s1s8s Fast Read cmd inst 7Ch support(no)

fr_inst_1s8s8s field 1s8s8s Fast Read cmd inst CCh support(no)

fr_inst_1s8d8d field 1s8d8d DTR Read cmd inst FDh support(no)

page_program_support_1s1s8s field 1s1s8s page program support (no)

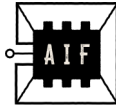
page_program_support_1s8s8s field 1s8s8s page program support (no)

reg_6_7_4 register

- Absolute Address: 0x704
- Base Offset: 0x704
- Size: 0x4

Bits Identifier	Access	Reset	Name
31:0	erase_reserved	r	0x0 —

erase_reserved field erase is not supported

**reg_6_9_3 register**

- Absolute Address: 0x900
- Base Offset: 0x900
- Size: 0x4

Bits	Identifier	Access	Reset	Name
4:0	reserved	r	0x0	—
5	enter_spi_supported	r	0x1	—
14:6	pgm_reserved	r	0x0	—
15	deep_pd_support	r	0x1	—
16	cfg_reg_load_support	r	0x1	—
17	cfg_reg_read_support	r	0x1	—
18	sts_reg_clr_support	r	0x1	—
19	sts_reg_read_support	r	0x1	—
20	sren_support	r	0x0	—
21	wren2_support	r	0x0	—
22	wren1_support	r	0x0	—
23	write_mem_linear	r	0x1	—
24	write_mem_wrapped	r	0x0	—
25	write_reg_linear	r	0x1	—
26	write_reg_wrapped	r	0x0	—
27	read_mem_linear	r	0x1	—
28	read_mem_wrapped	r	0x0	—
29	read_reg_linear	r	0x1	—
30	read_reg_wrapped	r	0x0	—
31	xspi_profile_2_support	r	0x1	—

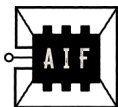
reserved field reserved

enter_spi_supported field enter SPI supported

pgm_reserved field program/erase not supported

deep_pd_support field Deep Power down support(yes)

cfg_reg_load_support field Cfg register load support(yes)



CONTENTS

cfg_reg_read_suport field Cfg register read support(yes)

sts_reg_clr_support field Status register clear support(yes)

sts_reg_read_support field Status register read support(yes)

sren_support field SREN support(no)

wren2_support field WREN2 support(no)

wren1_support field WREN1 support(no)

write_mem_linear field write memory linear support

write_mem_wrapped field write memory wrapped support

write_reg_linear field write register linear support

write_reg_wrapped field write register wrapped support

read_mem_linear field Read memory linear support

read_mem_wrapped field Read memory wrapped support

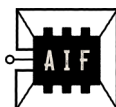
read_reg_linear field Read register linear support

read_reg_wrapped field Read register wrapped support

xspi_profile_2_support field xspi profile2 support

reg_6_9_4 register

- Absolute Address: 0x904
- Base Offset: 0x904
- Size: 0x4



Bits Identifier	Access	Reset	Name
6:2	cfg_bit_pattern_num_dymmy_cycl_required	r	0x2 —
11:7	num_dymmy_cycl_required	r	0x2 —

cfg_bit_pattern_num_dymmy_cycl_required field 200Mhz num cfg bit pattern to set num dummy cycl required TBD

num_dymmy_cycl_required field 200Mhz num dummy cycl required TBD

reg_6_9_5 register

- Absolute Address: 0x908
- Base Offset: 0x908
- Size: 0x4

Bits	Identifier	Access	Reset	Name
6:2	cfg_bit_pattern_num_dymmy_cycl_required_100	r	0x2	—
11:7	num_dymmy_cycl_required_100	r	0x2	—
16:12	cfg_bit_pattern_num_dymmy_cycl_required_133	r	0x2	—
21:17	num_dymmy_cycl_required_133	r	0x2	—
26:22	cfg_bit_pattern_num_dymmy_cycl_required_166	r	0x2	—
31:27	num_dymmy_cycl_required_166	r	0x2	—

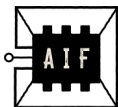
cfg_bit_pattern_num_dymmy_cycl_required_100 field 100Mhz num cfg bit pattern to set num dummy cycl required TBD

num_dymmy_cycl_required_100 field 100Mhz num dummy cycl required TBD

cfg_bit_pattern_num_dymmy_cycl_required_133 field 133Mhz num cfg bit pattern to set num dummy cycl required TBD

num_dymmy_cycl_required_133 field 133Mhz num dummy cycl required TBD

cfg_bit_pattern_num_dymmy_cycl_required_166 field 166Mhz num cfg bit pattern to set num dummy cycl required TBD



num_dymmy_cycl_required_166 field 166Mhz num dummy cycl required
TBD

reg_6_10_3 register

- Absolute Address: 0xA00
- Base Offset: 0xA00
- Size: 0x4

Bits Identifier	Access	Reset	Name
31:0	volatile_address	r	0x0 —

volatile_address field Address offset for volatile registers TBD

reg_6_10_4 register

- Absolute Address: 0xA04
- Base Offset: 0xA04
- Size: 0x4

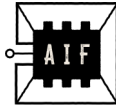
Bits Identifier	Access	Reset	Name
31:0	nonvolatile_address	r	0x0 —

nonvolatile_address field Address offset for non-volatile registers not supported

reg_6_10_5 register

- Absolute Address: 0xA08
- Base Offset: 0xA08
- Size: 0x4

Bits	Identifier	Access	Reset	Name
9:6	num_dummy_cycles_8d8d8d	r	0x0	—
13:10	num_dummy_cycles_8s8s8s	r	0x0	—
17:14	num_dummy_cycles_4d4d4d	r	0x0	—



Bits	Identifier	Access	Reset	Name
21:18	num_dummy_cycles_4s4s4s	r	0x0	—
25:22	num_dummy_cycles_2s2s2s	r	0xF	—
27:26	num_dummy_cycles_1s1s1s	r	0x0	—
29:28	num_addr_bytes	r	0x3	—
30	gen_reg_write_supported	r	0x1	—
31	gen_reg_read_supported	r	0x1	—

num_dummy_cycles_8d8d8d field Number of dummy cycles for read in 8d-8d-8d mode

num_dummy_cycles_8s8s8s field Number of dummy cycles for read in 8s-8s-8s mode

num_dummy_cycles_4d4d4d field Number of dummy cycles for read in 4d-4d-4d mode

num_dummy_cycles_4s4s4s field Number of dummy cycles for read in 4s-4s-4s mode

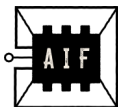
num_dummy_cycles_2s2s2s field Number of dummy cycles for read in 2s-2s-2s mode

num_dummy_cycles_1s1s1s field Number of dummy cycles for read in 1s-1s-1s mode

num_addr_bytes field Number address bytes(32bit)

gen_reg_write_supported field Generic Addressable Write Status/Control register command for volatile registers supported

gen_reg_read_supported field Generic Addressable Read Status/Control register command for volatile registers supported

**reg_6_10_6 register**

- Absolute Address: 0xA0C
- Base Offset: 0xA0C
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

reserved field No Non volatile registers

reg_6_10_7 register

- Absolute Address: 0xA10
- Base Offset: 0xA10
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

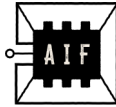
reserved field No Write in progress bit TBD

reg_6_10_8 register

- Absolute Address: 0xA14
- Base Offset: 0xA14
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

reserved field No Write enable bit TBD

**reg_6_10_9 register**

- Absolute Address: 0xA18
- Base Offset: 0xA18
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

reserved field No program Error bit

reg_6_10_10 register

- Absolute Address: 0xA1C
- Base Offset: 0xA1C
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

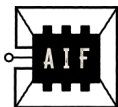
reserved field No Erase Error bit

reg_6_10_11 register

- Absolute Address: 0xA20
- Base Offset: 0xA20
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

reserved field Variable dummy cycles not supported

**reg_6_10_12 register**

- Absolute Address: 0xA24
- Base Offset: 0xA24
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

reserved field Variable dummy cycles (nvreg)not supported

reg_6_10_13 register

- Absolute Address: 0xA28
- Base Offset: 0xA28
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

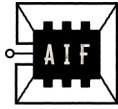
reserved field Variable dummy cycles not supported

reg_6_10_14 register

- Absolute Address: 0xA2C
- Base Offset: 0xA2C
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

reserved field Variable dummy cycles not supported

**reg_6_10_15 register**

- Absolute Address: 0xA30
- Base Offset: 0xA30
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

reserved field Variable dummy cycles not supported

reg_6_10_16 register

- Absolute Address: 0xA34
- Base Offset: 0xA34
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

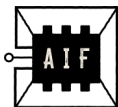
reserved field QPI mode not supported TBD

reg_6_10_17 register

- Absolute Address: 0xA38
- Base Offset: 0xA38
- Size: 0x4

Bits Identifier Access Reset Name		
31:0 reserved	r	0x0 —

reserved field QPI mode not supported TBD

**reg_6_10_18 register**

- Absolute Address: 0xA3C
- Base Offset: 0xA3C
- Size: 0x4

Bits	Identifier	Access	Reset	Name
23:16	reg_offset	r	0x0	—
26:24	bit_loc	r	0x0	—
27	local_addr_in_last_byte	r	0x0	—
28	addressable	r	0x1	—
30	polarity	r	0x0	—
31	octal_mode_enable	r	0x1	—

reg_offset field Reg Offset

bit_loc field Bit location in reg (TBD)

local_addr_in_last_byte field local address in last byte of address(yes)

addressable field Bit is addressable

polarity field Octal mode Enable polarity

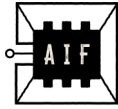
octal_mode_enable field STR/DDR Octal mode Enable Volatile

reg_6_10_19 register

- Absolute Address: 0xA40
- Base Offset: 0xA40
- Size: 0x4

Bits	Identifier	Access	Reset	Name
31:0	reserved	r	0x0	—

reserved field Octal mode Enable non Volatile (not supported)

**reg_6_10_20 register**

- Absolute Address: 0xA44
- Base Offset: 0xA44
- Size: 0x4

Bits	Identifier	Access	Reset	Name
23:16	reg_offset	r	0x0	—
26:24	bit_loc	r	0x0	—
27	local_addr_in_last_byte	r	0x0	—
28	addressable	r	0x1	—
30	polarity	r	0x0	—
31	ddr_mode_select_available	r	0x1	—

reg_offset field Reg Offset

bit_loc field Bit location in reg (TBD)

local_addr_in_last_byte field local address in last byte of address(yes)

addressable field Bit is addressable

polarity field polarity

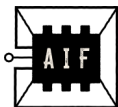
ddr_mode_select_available field SDR/DDR mode select volatile (supported)

reg_6_10_21 register

- Absolute Address: 0xA48
- Base Offset: 0xA48
- Size: 0x4

Bits	Identifier	Access	Reset	Name
31:0	reserved	r	0x0	—

reserved field SDR/DDR non Volatile (not supported)

**reg_6_10_22 register**

- Absolute Address: 0xA4C
- Base Offset: 0xA4C
- Size: 0x4

Bits	Identifier	Access	Reset	Name
23:16	reg_offset	r	0x0	—
26:24	bit_loc	r	0x0	—
27	local_addr_in_last_byte	r	0x0	—
28	addressable	r	0x1	—
30	polarity	r	0x0	—
31	octal_mode_enable	r	0x1	—

reg_offset field Reg Offset

bit_loc field Bit location in reg (TBD)

local_addr_in_last_byte field local address in last byte of address(yes)

addressable field Bit is addressable

polarity field Octal mode Enable polarity

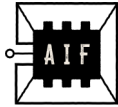
octal_mode_enable field STR Octal mode Enable Volatile Dup of 6.10.18?
TBD

reg_6_10_23 register

- Absolute Address: 0xA50
- Base Offset: 0xA50
- Size: 0x4

Bits	Identifier	Access	Reset	Name
31:0	reserved	r	0x0	—

reserved field SDR/DDR non Volatile (not supported)

**reg_6_10_24 register**

- Absolute Address: 0xA54
- Base Offset: 0xA54
- Size: 0x4

Bits	Identifier	Access	Reset	Name
23:16	reg_offset	r	0x0	—
26:24	bit_loc	r	0x0	—
27	local_addr_in_last_byte	r	0x0	—
28	addressable	r	0x1	—
30	polarity	r	0x0	—
31	octal_mode_enable	r	0x1	—

reg_offset field Reg Offset

bit_loc field Bit location in reg (TBD)

local_addr_in_last_byte field local address in last byte of address(yes)

addressable field Bit is addressable

polarity field Octal mode Enable polarity

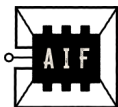
octal_mode_enable field DTR Octal mode Enable Volatile Dup of 6_10_18?
TBD

reg_6_10_25 register

- Absolute Address: 0xA58
- Base Offset: 0xA58
- Size: 0x4

Bits	Identifier	Access	Reset	Name
31:0	reserved	r	0x0	—

reserved field SDR/DDR non Volatile (not supported)

**reg_6_10_26 register**

- Absolute Address: 0xA5C
- Base Offset: 0xA5C
- Size: 0x4

Bits	Identifier	Access	Reset	Name
23:16	reg_offset	r	0x0	—
26:24	bit_loc	r	0x0	—
27	local_addr_in_last_byte	r	0x0	—
28	addressable	r	0x1	—
30	polarity	r	0x0	—
31	ddr_mode_select_available	r	0x1	—

reg_offset field Reg Offset

bit_loc field Bit location in reg (TBD)

local_addr_in_last_byte field local address in last byte of address(yes)

addressable field Bit is addressable

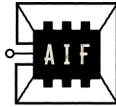
polarity field polarity

ddr_mode_select_available field Deep power down select volatile (supported)

reg_6_10_27 register

- Absolute Address: 0xA60
- Base Offset: 0xA60
- Size: 0x4

Bits	Identifier	Access	Reset	Name
23:16	reg_offset	r	0x0	—
26:24	bit_loc	r	0x0	—
27	local_addr_in_last_byte	r	0x0	—



Bits	Identifier	Access	Reset	Name
28	addressable	r	0x1	—
30	polarity	r	0x0	—
31	ddr_mode_select_available	r	0x1	—

reg_offset field Reg Offset

bit_loc field Bit location in reg (TBD)

local_addr_in_last_byte field local address in last byte of address(yes)

addressable field Bit is addressable

polarity field polarity

ddr_mode_select_available field Ultra Deep power down select volatile (supported)

reg_6_10_28 register

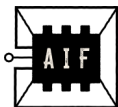
- Absolute Address: 0xA64
- Base Offset: 0xA64
- Size: 0x4

Bits	Identifier	Access	Reset	Name
23:16	reg_offset	r	0x0	—
26:24	bit_loc	r	0x0	—
27	local_addr_in_last_byte	r	0x0	—
28	addressable	r	0x1	—
31:30	num_bits	r	0x3	—

reg_offset field Reg Offset

bit_loc field Bit location in reg (TBD)

local_addr_in_last_byte field local address in last byte of address(yes)



addressable field Bit is addressable

num_bits field Output drive strength numbits(supported)

reg_6_10_29 register

- Absolute Address: 0xA68
- Base Offset: 0xA68
- Size: 0x4

Bits Identifier Access Reset Name			
31:0 reserved	r	0x0	—

reserved field non Volatile (not supported)

reg_6_11_3 register

- Absolute Address: 0xA6C
- Base Offset: 0xA6C
- Size: 0x4

Bits Identifier Access Reset Name			
31:0 offset	r	0x0	—

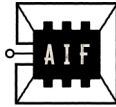
offset field Volatile reg offset TBD

reg_6_11_4 register

- Absolute Address: 0xA70
- Base Offset: 0xA70
- Size: 0x4

Bits Identifier Access Reset Name			
31:0 offset	r	0x0	—

offset field non Volatile reg offset TBD

**reg_6_11_5 register**

- Absolute Address: 0xB00
- Base Offset: 0xB00
- Size: 0x4

Bits	Identifier	Access	Reset	Name
2:0	reserved	r	0x0	—
10:3	wip_address	r	0x0	—
14:11	wip_bit_location	r	0x0	—
15	wip_address_byte_location	r	0x0	—
16	wip_polarity	r	0x0	—
17	wip_supported	r	0x0	—
22:18	dummy_cycles_non_volative	r	0x0	—
27:23	dummy_cycles_volative	r	0x0	—
29:28	address_bytes	r	0x0	—
30	write_ctrl_sts	r	0x1	—
31	read_ctrl_sts	r	0x1	—

reserved field Reserved

wip_address field Address of WIP Bit(not supported)

wip_bit_location field Bit location of WIP Bit(not supported)

wip_address_byte_location field Local address of WIP Bit(not supported)

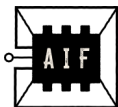
wip_polarity field WIP Polarity(not supported)

wip_supported field WIP (Device busy bit) (not supported)

dummy_cycles_non_volative field Dummy cycles for non-volatile reg read
TBD

dummy_cycles_volative field Dummy cycles for volatile reg read TBD

address_bytes field Address Bytes(6 bytes)



write_ctrl_sts field Generic addressable write ctrl/sts registers(yes)

read_ctrl_sts field Generic addressable read ctrl/sts registers(yes)

reg_6_11_6 register

- Absolute Address: 0xB04
- Base Offset: 0xB04
- Size: 0x4

Bits	Identifier	Access	Reset	Name
31:0	p_error	r	0x0	—

p_error field Program Error (Not Implemented)

reg_6_11_7 register

- Absolute Address: 0xB08
- Base Offset: 0xB08
- Size: 0x4

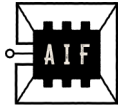
Bits	Identifier	Access	Reset	Name
31:0	e_error	r	0x0	—

e_error field Erase Error (Not Implemented)

reg_6_11_8 register

- Absolute Address: 0xB0C
- Base Offset: 0xB0C
- Size: 0x4

Bits	Identifier	Access	Reset	Name
30:0	reserved	r	0x0	—
31	supported	r	0x0	—



reserved field Reserved

supported field Variable number of dummy cycles volatile reg(Not Supported)

reg_6_11_9 register

- Absolute Address: 0xB10
- Base Offset: 0xB10
- Size: 0x4

Bits	Identifier	Access	Reset	Name
30:0	reserved	r	0x0	—
31	supported	r	0x0	—

reserved field Reserved

supported field Variable number of dummy cycles non-volatile reg(Not Supported)

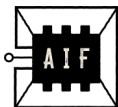
reg_6_11_10 register

- Absolute Address: 0xB14
- Base Offset: 0xB14
- Size: 0x4

Bits	Identifier	Access	Reset	Name
30:0	reserved	r	0x0	—
31	supported	r	0x0	—

reserved field Reserved

supported field Variable number of dummy cycles Bit pattern(Not Supported)



reg_6_11_11 register

- Absolute Address: 0xB18
- Base Offset: 0xB18
- Size: 0x4

Bits	Identifier	Access	Reset	Name
30:0	reserved	r	0x0	—
31	supported	r	0x0	—

reserved field Reserved

supported field Variable number of dummy cycles Bit pattern(Not Supported)

reg_6_11_12 register

- Absolute Address: 0xB1C
- Base Offset: 0xB1C
- Size: 0x4

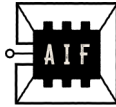
Bits	Identifier	Access	Reset	Name
30:0	reserved	r	0x0	—
31	supported	r	0x0	—

reserved field Reserved

supported field Variable number of dummy cycles Bit pattern(Not Supported)

reg_6_11_13 register

- Absolute Address: 0xB20
- Base Offset: 0xB20
- Size: 0x4



Bits	Identifier	Access	Reset	Name
24:17	addr	r	0x0	—
28:25	msb_bit_location	r	0x0	—
29	addr_not_in_last_byte	r	0x0	—
31:30	drive_strength_num_bits	r	0x3	—

addr field Address of register TBD (typo in spec?)

msb_bit_location field Bit location of MSB Bit TBD

addr_not_in_last_byte field addr in last_byte TBD

drive_strength_num_bits field output drive strength num_bits (volatile)

reg_6_11_14 register

- Absolute Address: 0xB24
- Base Offset: 0xB24
- Size: 0x4

Bits	Identifier	Access	Reset	Name
0	reserved	r	0x0	—

reserved field output drive strength nv (not implemented)

reg_6_11_15 register

- Absolute Address: 0xB28
- Base Offset: 0xB28
- Size: 0x4

Bits	Identifier	Access	Reset	Name
0	tbd	r	0x0	—

tbd field output drive strength bit pattern TBD

**reg_6_17_3 register**

- Absolute Address: 0xB2C
- Base Offset: 0xB2C
- Size: 0x4

Bits	Identifier	Access	Reset	Name
4	gram_spi	r	0x1	—
5	address_shift	r	0x0	—
11:8	md_address_format	r	0x4	—
19:16	md_address_write_access	r	0x0	—
23:20	md_address_read_access	r	0x0	—
27:24	md_active_die	r	0x0	—
31:28	md_address_offset	r	0x0	—

gram_spi field Gram SPI Mode TBD

address_shift field Address Shift TBD

md_address_format field Multi-Die Addressformat TBD

md_address_write_access field Multi-Die Addressable reg read access TBD

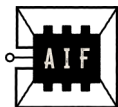
md_address_read_access field Multi-Die Addressable reg read access TBD

md_active_die field Active-Die Selection for Multi Die Package TBD

md_address_offset field Multi-Die Address offset for Addressable Register
TBD

reg_6_17_4 register

- Absolute Address: 0xB30
- Base Offset: 0xB30
- Size: 0x4



Bits	Identifier	Access	Reset	Name
7:0	write_opcode	r	0x71	—
15:8	read_opcode	r	0x65	—
20:16	dummy_cycles	r	0x8	—
23:21	dummy_cycles_override	r	0x0	—
27:24	write_enable	r	0x0	—
31:28	volatile_addr_offset	r	0x0	—

write_opcode field Write Opcode

read_opcode field Read Opcode

dummy_cycles field Dummy cycles

dummy_cycles_override field Dummy cycles override

write_enable field Write enable

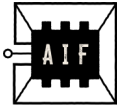
volatile_addr_offset field Volatile address offset TBD

reg_6_17_5 register

- Absolute Address: 0xB34
- Base Offset: 0xB34
- Size: 0x4

Bits	Identifier	Access	Reset	Name
7:0	write_opcode	r	0x71	—
15:8	read_opcode	r	0x65	—
20:16	dummy_cycles	r	0x8	—
23:21	dummy_cycles_override	r	0x0	—
27:24	write_enable	r	0x0	—
31:28	volatile_addr_offset	r	0x0	—

write_opcode field Write Opcode



CONTENTS

read_opcode field Read Opcode

dummy_cycles field Dummy cycles

dummy_cycles_override field Dummy cycles override

write_enable field Write enable

volatile_addr_offset field Volatile address offset TBD

reg_6_17_6 register

- Absolute Address: 0xB38
- Base Offset: 0xB38
- Size: 0x4

Bits	Identifier	Access	Reset	Name
3:0	statreg_bit0	r	0x1	—
11:8	statreg_access	r	0x1	—
14:12	dummy_cycles	r	0x2	—
23:16	aux1	r	0x0	—
31:24	aux2	r	0x5	—

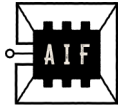
statreg_bit0 field SR bit 0 is busy flag

statreg_access field SR Access

dummy_cycles field Dymmy Cycles

aux1 field Aux 1 TBD SR address

aux2 field Aux 2



SCCR_Reg address map

- Absolute Address: 0x0
- Base Offset: 0x0
- Size: 0x38

Status and control Registers

Offset Identifier		Name
0x00	ID0	id0
0x08	ID1	ID1
0x10	CFG	—
0x18	xspi_status	—
0x20	xspi_control	—
0x28	xspi_rates	—
0x30	interrupt_status	—

ID0 register

- Absolute Address: 0x0
- Base Offset: 0x0
- Size: 0x8

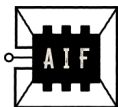
Bits Identifier Access Reset Name				
3:0	mgf_id	rw	0x2	—
15:4	devid0	rw	0x0	—

mgf_id field Manufacturer: TODO check how to obtain this number

devid0 field DEVID0 TODO Use as suitable

ID1 register

- Absolute Address: 0x8
- Base Offset: 0x8
- Size: 0x8



Bits	Identifier	Access	Reset	Name
3:0	dev_type	rw	0x0	—
15:4	devid1	rw	0x0	—

dev_type field Device Type set to hyperram

devid1 field TODO Use as appropriate

CFG register

- Absolute Address: 0x10
- Base Offset: 0x10
- Size: 0x8

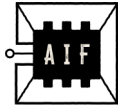
Bits	Identifier	Access	Reset	Name
1:0	BurstLength	rw	0x2	—
2	HybridBurstEnable	r	0x0	—
3	FixedLatency	r	0x1	—
7:4	InitialLatency	rw	0x8	—
11:8	Reserved	r	0x1	—
14:12	DriveStrength	r	0x0	—
15	DeepPowerDown	rw	0x0	—
16	BurstEnable	rw	0x0	—
17	UltraDeepPowerDown	rw	0x0	—

BurstEnable field Enable burst access. Applicable only to memory access. Register access are one at a time

xspi_status register

- Absolute Address: 0x18
- Base Offset: 0x18
- Size: 0x8

Bits	Identifier	Access	Reset	Name
0	wip	r	0x0	—



wip field write in progress

xspi_control register

- Absolute Address: 0x20
- Base Offset: 0x20
- Size: 0x8

Bits Identifier		Access Reset Name		
0	use_xspi_clk	rw	0x0	—
1	interrupt_enable	rw	0x0	—

use_xspi_clk field Use xspi clock as system clock

interrupt_enable field Use xspi clock as system clock

xspi_rates register

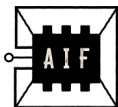
- Absolute Address: 0x28
- Base Offset: 0x28
- Size: 0x8

Bits	Identifier	Access	Reset	Name
7:0	cmd_rate	rw	0x0	—
15:8	addr_rate	rw	0x0	—
23:16	data_rate	rw	0x0	—

cmd_rate field CMD Rate

addr_rate field CMD Rate

data_rate field CMD Rate

**interrupt_status register**

- Absolute Address: 0x30
- Base Offset: 0x30
- Size: 0x8

Bits Identifier		Access Reset Name	
1:0	axi_resp	r, rclr	0x0 —
2	read_underflow	r, rclr	0x0 —
3	write_overflow	r, rclr	0x0 —

axi_resp field AXI Resp Error

read_underflow field Read FIFO Underflow Error

write_overflow field Write FIFO Overflow Error

Steps for running verification

Note

Arrow direction is from Initiator to Target.

Do we want the client to use the chiplet interface to preload a program in the IMEM of cpu subsystem.

Power Domains

1. Power switch is on VDD.
2. Isolation cells are on output ports
3. Logic outside the powerdomain blocks is

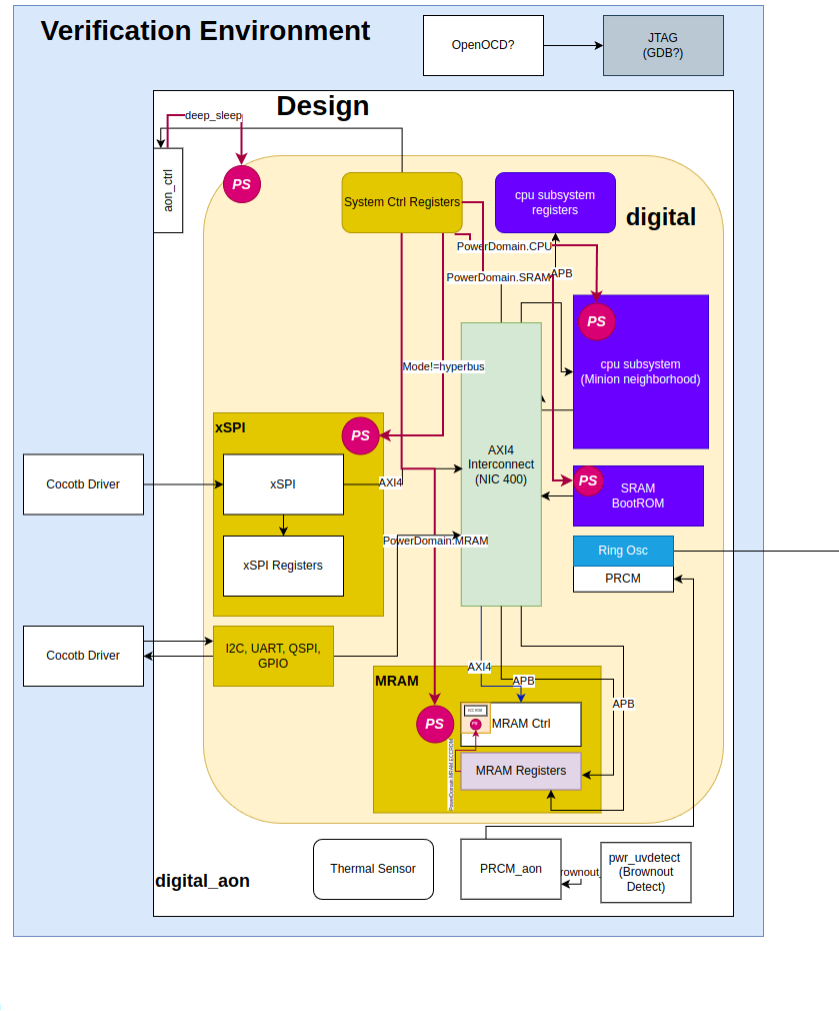


Figure 10: Verification Environment

- `git clone git@gitlab.veevx.net:erbium/hdl-et.git`
- `cd hdl-et/`
- `make -C tb setup`
- `uv sync`
- `source .venv/bin/activate`
- `make <MODULE=testname>`